

ADVANCED OPTICAL MATERIALS

Supporting Information

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Organic Solvent and Surfactant Free Fluorescent Organic
Nanoparticles by Laser Ablation of Aggregation-Induced
Enhanced Emission Dyes

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Experimental Section

Synthesis of 4-ethoxycarbonylbenzaldehyde

Terephthalaldehydic acid (5.5 g, 36.6 mmol), sulfuric acid (0.91 mL), and ethanol (166 mL) were added to a 250 mL round bottomed flask and then refluxed for 18 hours. Then the reaction mixture was cooled to RT and the solvent was evaporated. The remained mixture was dissolved in ethyl acetate (100 mL) and was washed with saturated sodium bicarbonate (200 mL $\times$ 2), brine (200 mL $\times$ 2), and distilled water (200 mL $\times$ 2). The organic layer was dried with sodium sulfate and the solvent was evaporated to obtain a colorless liquid (92% yield). $^1\text{H-NMR}$ (500 MHz, CDCl_3), δ 10.11 (s, 1H), 8.20 (d, 2H, $J=8$ Hz), 7.95 (d, 2H, $J=8$ Hz), 4.42 (q, 2H, $J=7.5$ Hz), 1.43 (t, 3H, $J=7.5$ Hz).

Synthesis of diethyl 4,4'-((1Z,1'Z)-1,4-phenylenebis(2-cyanoethene-2,1-diyl))dibenzoate (DCEtDCS)

4-ethoxycarbonylbenzaldehyde (1.7 g, 9.5 mmol), p-xylene dicyanide (0.5 g, 3.2 mmol), and tert-butyl alcohol (20 mL) were added to a 50 mL round bottomed flask and then were heated to 60 °C. Tetrabutylammonium hydroxide (1M solution in methanol, 1 mL) was added

dropwise into the reaction mixture and the mixture was stirred at 60 °C overnight. The resulting precipitate was filtered and washed with methanol. A pale-yellow powder (87% yield) was obtained by recrystallization from hot methanol solution. ¹H-NMR (500 MHz, DMSO-d₆), δ 8.11 (d, 4H, J=8.5 Hz), 8.05 (d, 4H, J=8.5 Hz), 7.97 (s, 4 H), 4.35 (q, 4H, J=7 Hz), 1.34 (t, 6H, J=7 Hz).

Cell culture and cytotoxicity assays

Human microglia cells were obtained as a gift from Dr S.Mahajan (University at Buffalo). These cells were grown in glass bottom dishes (Mattek) and cultured in Advanced DMEM (ThermoFisher Scientific), supplemented with 2.5% fetal calf serum (ThermoFisher Scientific), 1% glutamax, 1% Antibiotic Antimycotic Solution (ThermoFisher Scientific) at 37°C in a humidified atmosphere with 5% CO₂.

For the MTT Assay, cells were placed in 96-well plates and incubated with the pristine and chitosan coated FONs at concentration ranging from 2 to 20 μM for either 3 h or 24 h at 37 °C. Next, cells were washed with PBS and incubated with 0.5 mg/mL of the MTT solution for 3 h, followed by the addition of dimethylsulfoxide. Plates were placed in the microplate spectrophotometer system (Opsys MR, Dynex technologies) and absorbance was recorded at 490 nm. Results were analyzed with the Revelation Quicklink software and are presented as the percentage of the control values.

To estimate effect of nanoparticles on cell division, mitotic cells were counted per field of view of 20× objective. Obtained numbers were then averaged from counting in 10 fields of view.

FON concentration calculation

We followed a reported method^[1] to calculate DCETDCS FON concentration. We calculated the concentration from the laser ablated FON (0.125 mg/mL) suspension. The density of the FON was estimated as 1 g/cm³ and the size of FON was 1.6 nm on AFM image.

The molar concentration was determined,

$$\begin{aligned}
 & \frac{\text{moles of particles}}{\text{volume (L)}} \\
 & \left(\frac{\text{total volume of FONs}}{\text{average volume of FON}} \right) / \text{Avogadro's number} \\
 & = \frac{1 \times 10^{-3} \text{ L}}{\left(\frac{\left(0.125 \times 10^{-3} \text{ g} / \frac{1 \text{ g}}{\text{mL}} \right)}{\frac{4}{3} \pi (0.8 \times 10^{-7})^3 \text{ mL}} \right) / 6.02 \times 10^{23} \text{ mol}^{-1}} / 1 \times 10^{-3} \text{ L} \\
 & = 9.7 \times 10^{-5} \text{ M}
 \end{aligned}$$

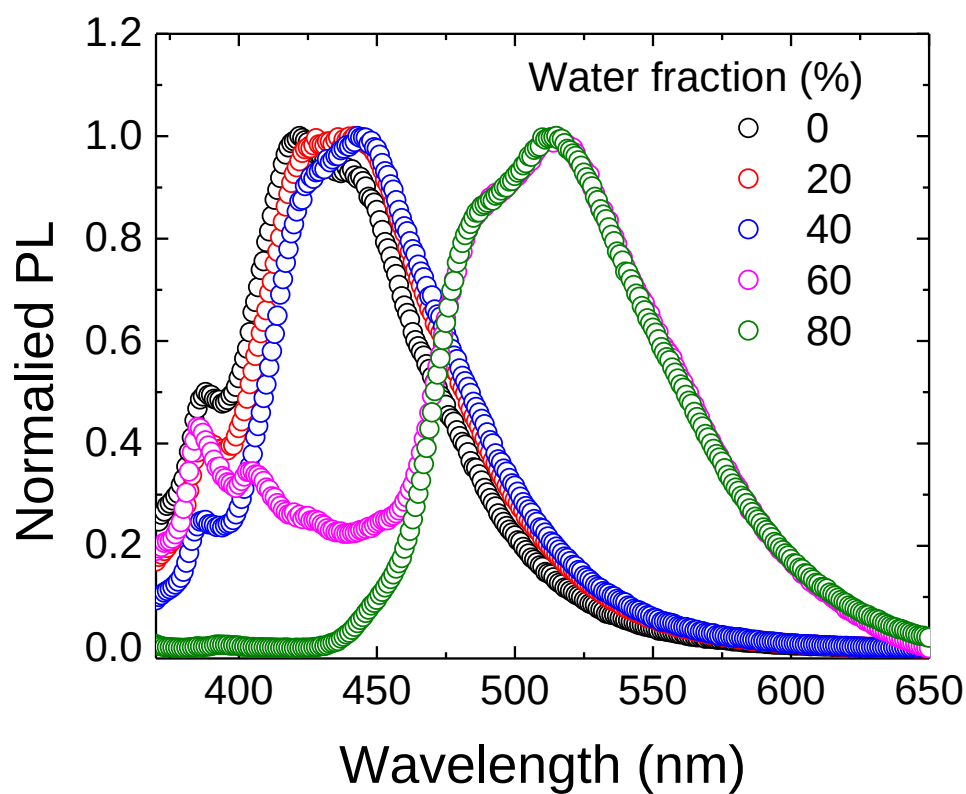


Figure S1. Normalized PL spectra of DCEtDCS in THF/water mixed solvent.

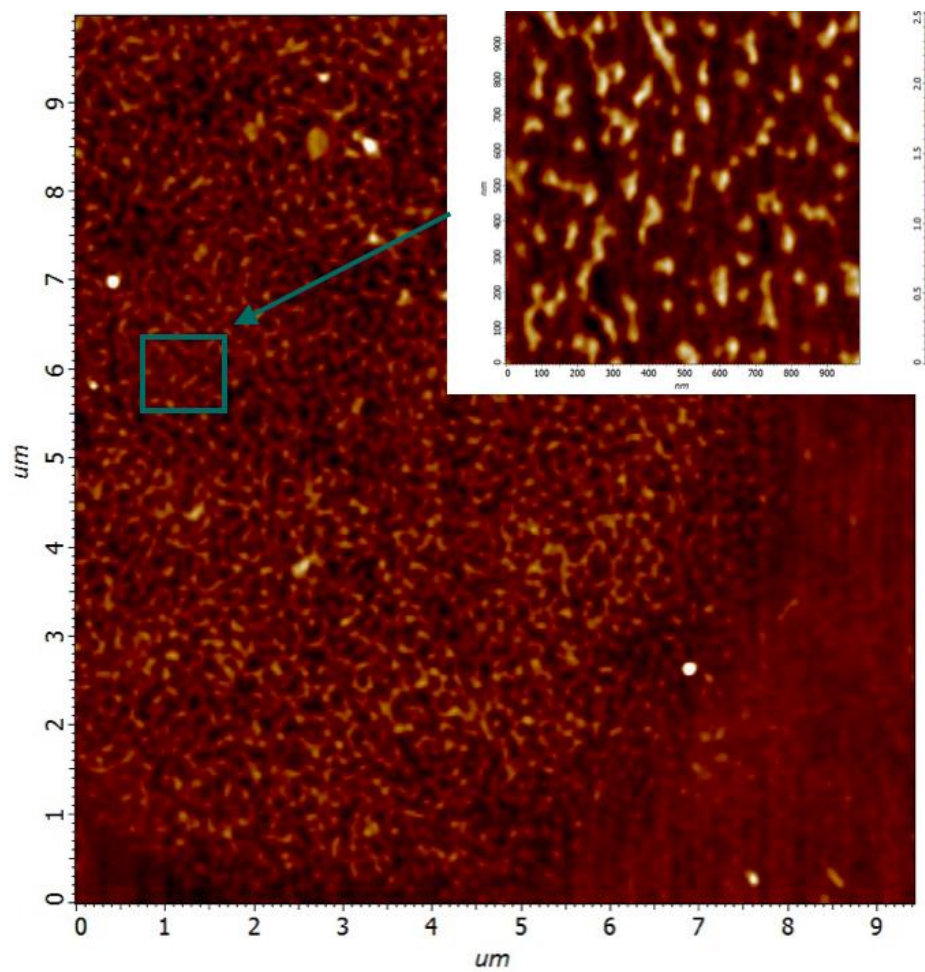


Figure S2. AFM image of laser ablated DCetDCS FONs

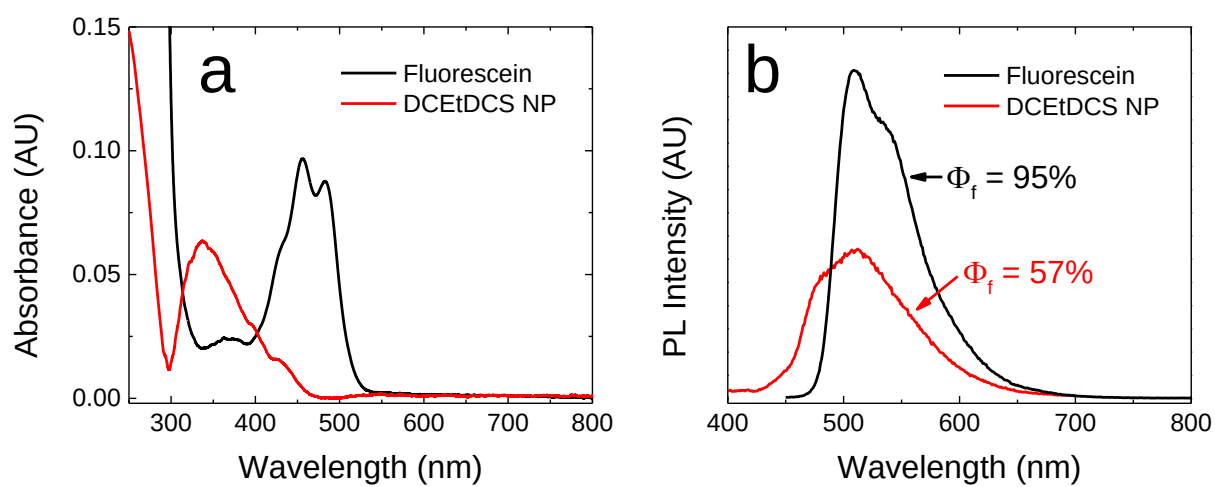


Figure S3. Determination of the quantum yield for laser ablated DCETDCS FON using fluorescein as a standard reference. (a) The absorbances of fluorescein and laser ablated DCETDCS FON (b) The measured PLQY of laser ablated DCETDCS FON (57%) was determined relative to the standard reference fluorescein (PLQY: 95% in 0.1M aq. NaOH).

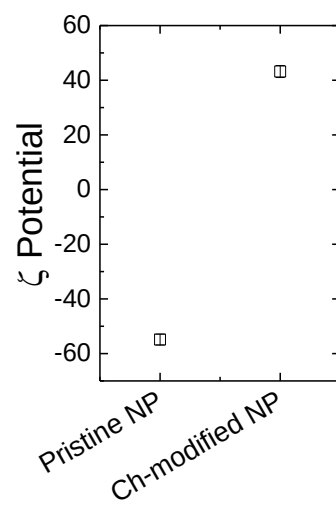


Figure S4. Surface charge (zeta potential) of pristine and chitosan (Ch) modified AIE FON.

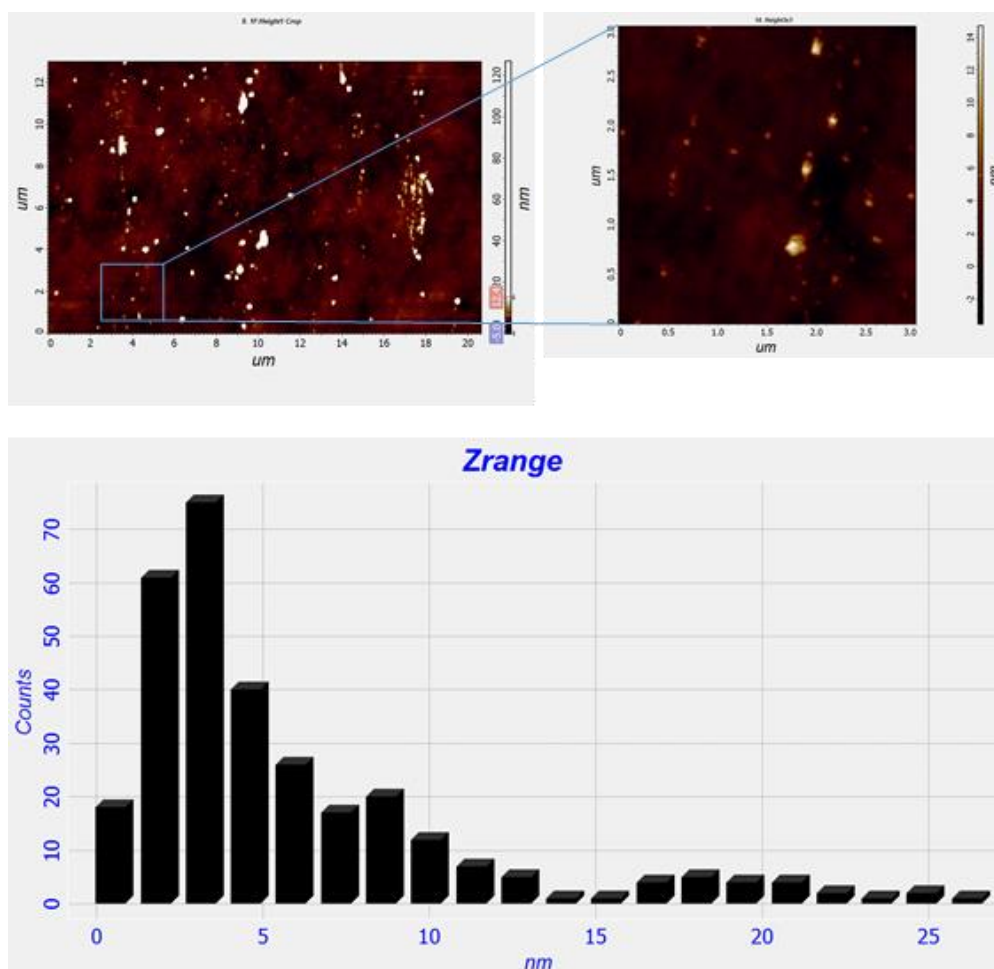


Figure S5. AFM image and height distribution of chitosan modified DCEtDCS FONs.

Reference

- [1] B. Chen, G. Feng, B. He, C. Goh, S. Xu, G. Ramos-Ortiz, L. Aparicio-Ixta, J. Zhou, L. Ng, Z. Zhao, *Small* **2016**, 12, 782.